Option C – The Physics of Sports

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)	Any valid moment of inertia equation used i.e. $I = kmr^2$ (1) Valid conclusion based on value of moment of inertia of C with all values for A and B also considered (1)			2	2	1	
		(ii)	Angular momentum $L = I\omega$ used (1) Angular momentum = 22 [kg m^2 s^{-1}] (1)	1	1		2	1	
	(b)	(i)	Angular acceleration = $\frac{\text{change in angular velocity}}{\text{time used}}$ (1) Angular acceleration = 1 256 [rad s^{-2}] (1) Moment of inertia = 7.35×10^{-5} [kg m^2] (1) Torque = 0.099 [N m] (1)	1	1 1 1		4	3	
		(ii)	Rotational kinetic energy $= \frac{1}{2}I\omega^2$ used (1) Rotational KE = 2.5 [J] (**ecf** on I and angular velocity) (1)	1	1		2	2	
		(iii)	Ball has linear and rotational KE or recall $F = \frac{mv-mu}{t}$ (1) Energy has to be dissipated or the glove will <u>increase time</u> of contact when being caught (1) Reduce the effect of friction on the hand (1)	1	1 1		3		

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	(c)	(i)	The velocity of the ball has reduced so KE is decreased as well as potential energy (1) By conservation of energy – energy lost against air resistance (1) **Alternative:** Both horizontal components of velocity calculated i.e. 36 cos3° and 33 cos7° (= 35.95 m s^{-1} and 32.75 m s^{-1} respectively) (1) Horizontal component of velocity has decreased so air resistance is present (1)		2		2		
		(ii)	Both vertical components calculated correctly i.e. 36sin3° = 1.89 [m s^{-1}] and 33sin7° = 4.02 [m s^{-1}] (1) Substitute values into a valid equation of motion e.g. $\frac{v^2 - u^2}{2ax}$ i.e. $\frac{4.02^2 - 1.89^2}{2 \times 0.75}$ (1) Valid conclusion (1) e.g. acceleration = 8.39 [m s^{-2}] so not equal to g **or** Height = 0.64 [m] (with g = 9.81m s^{-2}) so not equal to 0.75 [m] **Or** vertical velocity = 4.22 [m s^{-1}] (with g = 9.81 m s^{-2}) so not equal to 4.02 [m s^{-1}]			3	3	3	
		(iii)	A lift force [is created by a spinning ball] (1) can be implied Due to pressure or velocity <u>difference</u> caused by the spinning or equivalent explanation (1)	2			2		
			Question 9 total	**6**	**9**	**5**	**20**	**10**	**0**